

Could a transformer recover predictive state?

Belief-geometry primer · before Notebook 2 · 8–10 minutes

ILIAD Intensive · Computational Mechanics

The discovery puzzle

A transformer is trained only on next-token loss. Why look for anything richer than its next-token probabilities?

History 01

Belief (0, 0, 1)

Next token (1/2, 1/2)

History 10

Belief (1/2, 1/2, 0)

Next token (1/2, 1/2)

The objective scores one step at a time; good computation can still depend on distinctions that matter later.

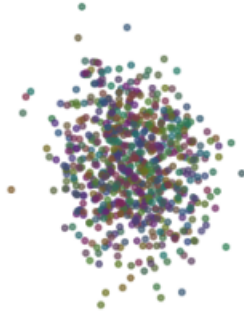
Where can we look without changing the model?

BELIEF GEOMETRY RECOVERY — color = exact belief; position = probe prediction

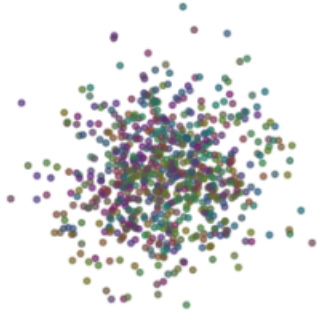
GROUND TRUTH
held-out subtrees



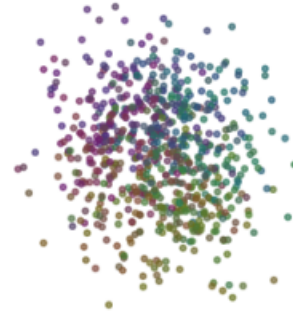
INITIALIZATION
MSE=0.0127



EARLY
MSE=0.0120



MIDDLE
MSE=0.0072



LATE
MSE=0.0004



Late recovery means one affine map extrapolates to prefix subtrees excluded from fitting.

A probe is an external diagnostic fitted after training—not part of the transformer's forward pass.

Climb an evidence ladder, not a picture gallery

- 1 Fit:** can a simple diagnostic recover belief coordinates? starting clue
- 2 Generalize:** does it work on held-out histories? structure
- 3 Discriminate:** does it beat simpler next-token or covariate explanations? specificity
- 4 Intervene:** does changing the decoded direction change computation as predicted? causal use

Predict the controls before seeing results

Held-out histories

Does one map extrapolate?

Shuffled pairing

Could high dimension fake the fit?

Training checkpoints

Does structure emerge with learning?

Simple baselines

Would token, length, or next-token probabilities suffice?

PROVENANCE RULE

Keep **teaching simulation**, **reported paper result**, and **our inference** visibly separate.

Your route: question first, reveal last

hypothesis



paired data



probe



controls



matched-prediction test

OPEN NOTEBOOK 2

Your job is to design the test and predict its failure modes before the reported result is shown.